

Multi Agentic Workflow for student Attendance Recognition and Real-time Monitoring LMS

An AI-Powered Learning Management System

BUG REPORT DOCUMENT

A bug is a defect or unexpected behavior in software that causes an application to produce incorrect results, crash, reduce performance, or create security risks. During testing, each identified bug was documented, reproduced, analyzed, fixed, and retested to verify the resolution. This report summarizes eight bugs identified across the Multi Agentic Workflow for student Attendance Recognition platform, spanning access control, data handling, AI-agent behavior, and system performance.

Summary of Reported Bugs

Bug ID	Title	Status
BUG-001	Unauthorized Dashboard Access	Closed
BUG-002	Missing Course Grades Display	Closed
BUG-003	Duplicate AI Study Plan Suggestions	Closed
BUG-004	Incorrect Attendance Percentage	Closed
BUG-005	AI Response Timeout on Large Syllabus	Closed
BUG-006	Unresponsive Mobile Sidebar	Closed
BUG-007	Database Lock on Concurrent RAG Queries	Closed
BUG-008	Missing Announcement Emails	Closed

Detailed Bug Reports

BUG- 001 — Unauthorized Dashboard Access

Severity: **Critical** Status: **Closed**

Problem: Students were occasionally able to navigate to the Teacher Dashboard URL directly without being redirected.

Root Cause: Role-based route guarding on the React frontend was not properly checking the user's role token on initial page load.

Resolution: Implemented a higher-order component (HOC) for protected routes that strictly validates the user's role against the backend JWT token before rendering the dashboard.

Verification: Attempted to access /teacher-dashboard while logged in as a student; the system successfully redirected to the unauthorized access page.

BUG- 002 — Missing Course Grades Display

Severity: **Medium** Status: **Closed**

Problem: The Student Dashboard displayed a blank screen instead of the grades table when a course had no graded assignments yet.

Root Cause: The frontend did not account for null or empty array responses from the backend grades API, causing a mapping error in React.

Resolution: Added conditional rendering to display a "No grades posted yet" fallback message when the data array is empty.

Verification: Viewed a newly created course as a student and confirmed the fallback message appeared instead of crashing the UI.

BUG-003 — Duplicate AI Study Plan Suggestions

Severity: **High** Status: **Closed**

Problem: The AI Agent generated duplicate tasks and study sessions when creating a personalized study plan for a student.

Root Cause: The prompt engineering lacked explicit instructions to prevent overlapping time slots, and the backend did not filter out duplicate generated entries.

Resolution: Updated the LLM system prompt to enforce unique task generation and implemented a deduplication function in the Python backend before saving to the database.

Verification: Generated multiple study plans and confirmed all tasks and time slots were unique.

BUG-004 — Incorrect Attendance Percentage

Severity: **High** Status: **Closed**

Problem: The overall attendance percentage for students was calculating to over 100% in certain edge cases.

Root Cause: Excused absences were being counted as both "attended" and "total days," skewing the mathematical ratio.

Resolution: Updated the calculation logic in academic.py to correctly exclude excused absences from the total required days denominator.

Verification: The calculated attendance matched manual verification for students with excused absences.

BUG-005 — AI Response Timeout on Large Syllabus

Severity: **Medium** Status: **Closed**

Problem: The AI Module occasionally timed out when asked to summarize or generate questions from a large PDF syllabus.

Root Cause: The text chunking process for the syllabus was too large, causing the LLM API to exceed its processing time limit.

Resolution: Added a background task queue to handle large document processing asynchronously and improved the RAG (Retrieval-Augmented Generation) chunking strategy.

Verification: Uploaded a 50-page syllabus and confirmed the UI displayed a "Processing" state until the summary was successfully returned.

BUG-006 — Unresponsive Mobile Sidebar

Severity: **Low** Status: **Closed**

Problem: The main navigation sidebar could not be closed on mobile devices after opening.

Root Cause: The click-away listener on the frontend was conflicting with the sidebar's toggle state.

Resolution: Refactored the sidebar state management in Tailwind/React to properly detect outside clicks and close the menu.

Verification: Tested the UI on mobile viewport dimensions and confirmed the sidebar opens and closes as expected.

BUG-007 — Database Lock on Concurrent RAG Queries

Severity: **Critical** Status: **Closed**

Problem: Simultaneous queries to the AI assistant by multiple students caused the database to lock and throw errors.

Root Cause: SQLite database connections were not properly configured for concurrent read/write operations during heavy AI interactions.

Resolution: Enabled Write-Ahead Logging (WAL) mode for the database and implemented connection pooling in FastAPI.

Verification: Simulated 50 concurrent AI requests using a load testing script; all requests resolved without database locks.

BUG-008 — Missing Announcement Emails

Severity: **Medium** Status: **Closed**

Problem: Email notifications were not being sent when a teacher posted a new announcement.

Root Cause: The email service function was failing silently because it was not awaiting the SMTP connection promise.

Resolution: Updated email_service.py to properly handle asynchronous SMTP requests and added error logging for failed dispatches.

Verification: Created a test announcement and confirmed the notification email was received in the test inbox.

Conclusion

All eight identified bugs were resolved and verified through regression testing. The Multi Agentic Workflow for student Attendance Recognition framework now demonstrates reliable role-based access control, accurate data calculation, stable AI-assisted study planning, resilient concurrent database handling, and robust notification delivery.